

RISETECH INTERNSHIP 2024

Embedded Systems

Project Report

Precision Agriculture

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Abstract:

This project presents the design, development, and deployment of a low-power, long-range wireless sensor network (WSN) for precision agriculture, utilizing STM32 microcontrollers and LoRa technology. The system aims to address the challenges of limited coverage and battery life faced by traditional precision agriculture solutions. By leveraging LoRa's capability to transmit data over distances up to 10 kilometers, this project provides a scalable and cost-effective solution for monitoring critical environmental parameters, including soil moisture, temperature, humidity, and light intensity, in remote agricultural fields.

The sensor nodes, based on the **STM32WL55JC1** microcontroller, integrate multiple sensors such as the **DHT22**, **BH1750**, and **HW-390** to collect and transmit data via the **LoRaWAN protocol**. A Dragino gateway forwards the data to The Things Stack network server, where it is processed and subsequently visualized on the ThingsBoard platform, enabling real-time data-driven decision-making for farmers. The system emphasizes low power consumption, with the microcontroller entering stop mode when idle, **drawing an average current of 7-12 mA**. Powered by a 4000mAh 7.4V LiPo battery, the sensor nodes achieve a **battery life of approximately 14 to 24 days**, with potential future enhancements including solar or energy harvesting technologies for self-sufficiency.

The implementation demonstrates minimal deviation between sensor readings and actual environmental conditions, validating the accuracy and reliability of the system. This innovation in precision agriculture empowers farmers to optimize resource management, improve crop yields, and promote sustainable farming practices, making it a promising solution for the future of agriculture.

Introduction:

In modern agriculture, the efficient management of essential resources such as water, light, temperature, and soil nutrients is critical for maximizing crop yields and ensuring the sustainability of farming practices. As the demand for higher agricultural productivity increases, farmers face significant challenges in optimizing resource use across large and often remote agricultural fields. Traditional precision agriculture solutions, which rely on short-range wireless communication technologies like Wi-Fi or Bluetooth, often struggle to provide adequate coverage. These technologies typically cover distances ranging from a few dozen meters to a few hundred meters, making them insufficient for vast agricultural deployments where the fields may stretch over kilometers.

Moreover, the battery life of sensor nodes, which are responsible for collecting and transmitting data, presents another significant limitation. In large-scale agricultural setups, sensor nodes may be dispersed over wide areas, making frequent maintenance tasks such as replacing or recharging batteries impractical and costly. The need for a reliable, long-range communication solution with extended battery life is evident.

To overcome these challenges, this project focuses on the design, development, and deployment of a low-power, long-range wireless sensor network (WSN) that can monitor critical environmental parameters across extensive agricultural fields. By leveraging advanced microcontroller technology and long-range communication protocols, this project

aims to create a scalable, cost-effective solution that provides farmers with real-time insights into their crops, enabling more informed decision-making and improved resource management.

System Design and Architecture:

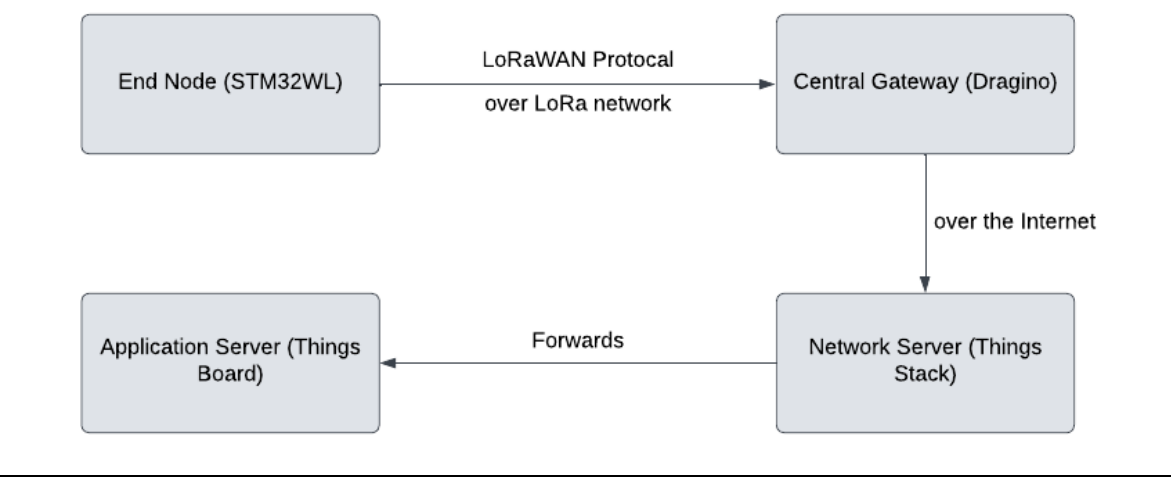


Figure 1: System Block Diagram



Figure 2: Network Server

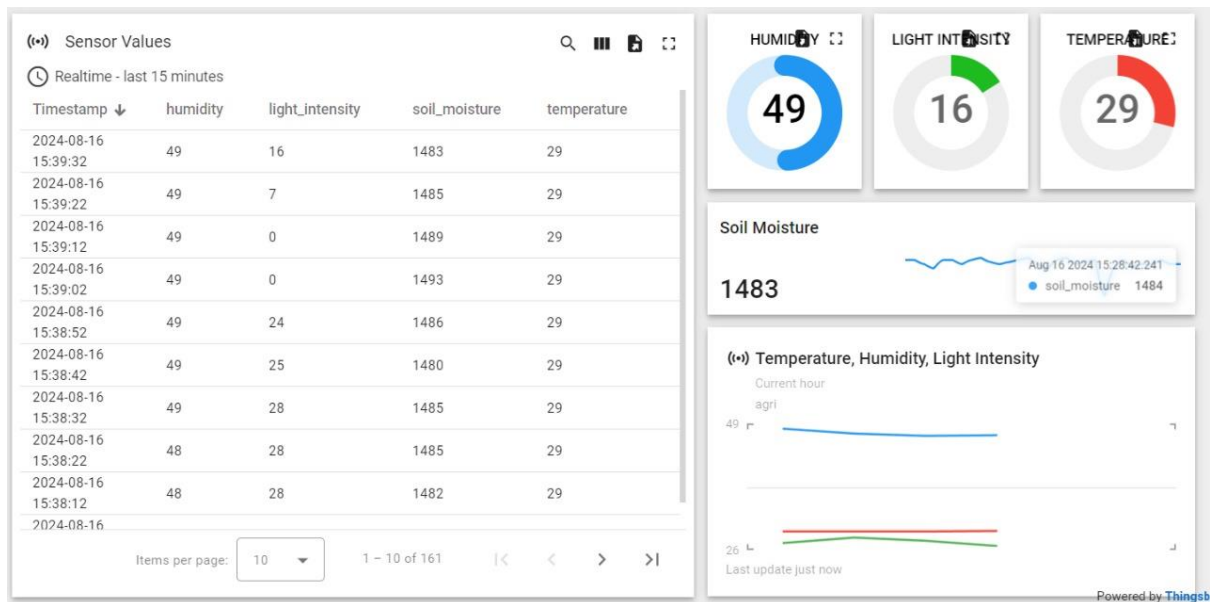


Figure 3: Application Server

Overview of the System Architecture:

- **Sensor Node Layer:** Consists of the STM32WL55JC1 microcontroller, sensors (DHT22, BH1750, HW-390), power management, and LoRaWAN communication.
- **Network Layer:** LoRaWAN gateway (e.g., Dragino) connects sensor nodes to the internet.
- **Network Server Layer:** The network server (e.g., The Things Stack) manages data decoding, processing, and routing.
- **Application Layer:** Application server (e.g., ThingsBoard) provides data visualization and analytics for users.

NOTE: Refer to the manual provided for Dragino, to connect your gateway/dragino onto a network server like things stack.

HARDWARE	DOCUMENTATION/REFERENCE LINK
STM32WL55 JC1	https://www.st.com/resource/en/user_manual/um2592-stm32wl-nucleo64-board-mb1389-stmicroelectronics.pdf
DHT22 (DIGITAL)	https://www.sparkfun.com/datasheets/Sensors/Temperature/DHT22.pdf
BH1750 (DIGITAL I2C)	https://www.handsontec.com/dataspecs/sensor/BH1750%20Light%20Sensor.pdf
HW-390 (ANALOG)	https://how2electronics.com/interface-capacitive-soil-moisture-sensor-arduino/

DRAGINO HP0D 1.3

<http://wiki.dragino.com/xwiki/bin/view/Main/User%20Manual%20for%20All%20Gateway%20models/HP0D/>

Software Components:

Firmware on Sensor Nodes: The firmware for the sensor node is structured to operate efficiently using **FreeRTOS**, which enables the concurrent execution of multiple tasks with different priorities. The **startLorawan** function is assigned a **higher priority** than the **StartDeadReckoning** function, which is responsible for acquiring sensor data. This priority setting ensures that the LoRaWAN communication, which is **time-sensitive and requires significant CPU time**, is consistently handled without delays or interruptions, maintaining reliable data transmission to the gateway. The LoRaWAN stack processes critical communication tasks, including joining the network, managing uplinks and downlinks, and adhering to specific timing constraints defined by the LoRaWAN protocol.

In contrast, the **StartDeadReckoning function**, which runs with a lower priority, **gathers environmental data from various sensors**. The DHT22 sensor, which measures temperature and humidity, operates based on specific timing requirements; for instance, it involves precise delays during data transmission to and from the microcontroller. This is managed by a custom **microDelay** function, ensuring accurate communication with the DHT22. The BH1750 light sensor is interfaced via the I2C protocol, allowing the sensor to continuously measure light intensity in lux and transmit this data to the microcontroller. The HW-390 soil moisture sensor is an analog sensor, and its output is read using the ADC (Analog-to-Digital Converter) of the microcontroller, which converts the analog voltage level corresponding to soil moisture into a digital value. This design allows the microcontroller to efficiently manage sensor readings while ensuring that the time-critical LoRaWAN communication takes precedence, optimizing the performance of the sensor node in a low-power, long-range wireless sensor network.

Important Files within Firmware: Should be configured accordingly.

FILES	DESCRIPTION
SYS_CONF.H	Applicative configuration, e.g. debug, trace, low power, sensors.
SE-IDENTITY.H	Sets application/network sessions keys etc.
FREERTOSCONFIG.H	Configuration for FreeRTOS e.g. tick, include/exclude API functions like taskRemove etc.
LORA_APP.H	Sets active region, transmission power, TX duty cycle, class of device etc.
LORAWAN_CONF.H	Sets LoRaWAN version e.g. 1.0.3.

Network Server Configuration (The Things Network/Stack):

Gateway Settings:

- **Gateway EUI:** Ensure the Gateway EUI in The Things Network (TTN) matches the one on the Dragino gateway.
- **Frequency Band:** Configure the same frequency band (e.g., EU868, US915) in both TTN and the Dragino gateway.

Application/End Device Settings:

- **LoRaWAN Version:** Match the LoRaWAN version in TTN with the end device firmware.
- **AppEUI and DevEUI:** The AppEUI and DevEUI in TTN must match those in the end device firmware.
- **Session Keys:** Ensure the Application Session Key (AppSKey) and Network Session Key (NwSKey) are identical in both TTN and the end device.

Implementation:

The proposed sensor network is built using the **STM32WL55JC1** microcontroller, which integrates both sensing and communication capabilities. The system is designed to monitor various environmental parameters, including **soil moisture, temperature, humidity, and light intensity**. These parameters are measured using different sensors: the **DHT22** for temperature and humidity, the **BH1750** for light intensity, and the **HW-390** for soil moisture.

Data collected by the sensor node is transmitted using the LoRaWAN protocol, leveraging the long-range capabilities of LoRa technology. LoRa is known for its ability to cover distances up to **10 kilometers in rural areas**, making it ideal for large-scale agricultural deployments. The data is sent over a LoRa network to a central gateway, specifically a **Dragino** device in this implementation. The gateway then forwards the data to a **network server**, The **Things Stack**, where the payloads are decoded and processed.

Once the data is processed, it is sent to an **application server, ThingsBoard**, for visualization and analysis. This setup enables farmers to access accurate and timely data, allowing them to make informed decisions that optimize crop management practices.

A critical aspect of this project is ensuring low power consumption to extend the battery life of the sensor nodes. The STM32WL55JC1 microcontroller is configured to enter stop mode when idle, significantly reducing power consumption. This approach enhances the viability of the system for long-term deployment in battery-operated environments.

By leveraging the advanced capabilities of STM32 microcontrollers and LoRa technology, this project successfully addresses the limitations of existing precision agriculture solutions. The resulting wireless sensor network is not only capable of providing robust and reliable communication in rural and remote areas but also promotes more sustainable and efficient agricultural practices.

Results and Analysis:

It was observed that the sensor readings, including temperature, humidity, and light intensity, showed minimal deviation from the actual environmental values. Specifically, the DHT22 sensor exhibited a difference of around 2% for temperature and 3% for humidity, the HW-390 soil moisture sensor showed a variation of approximately 4%, and the BH1750 light

intensity sensor had a discrepancy of about 1.5%. These differences are within acceptable limits for precision agriculture applications.

The entire circuitry draws an average current of around **7-12 mA**, and with a **4000mAh 7.4V LiPo battery**, the system is expected to operate for approximately **17.5 days** before needing a recharge. This battery life estimation highlights the efficiency of the low-power design, making it suitable for extended deployments in remote agricultural fields.

In the future, integrating solar panels or other energy harvesting methods could make the sensor nodes entirely self-sufficient, reducing or even eliminating the need for regular maintenance and battery replacement.

Conclusion:

This project successfully demonstrates the design, development, and deployment of a low-power, long-range wireless sensor network (WSN) tailored for precision agriculture. By leveraging STM32 microcontrollers and LoRa technology, the system addresses critical challenges in modern agriculture, such as limited wireless range.

The sensor nodes, built around the STM32WL55JC1 microcontroller, efficiently monitor key environmental parameters, including soil moisture, temperature, humidity, and light intensity. The data is transmitted using the LoRaWAN protocol to a Dragino gateway, which then forwards it to The Things Stack network server for processing. The integration with ThingsBoard allows for real-time visualization and analysis, providing farmers with actionable insights to optimize their crop management practices.

Throughout the project, a strong emphasis was placed on low power consumption, with the microcontroller entering stop mode when idle to extend battery life. The system's current draw, averaging between 7-12 mA, enables continuous operation for approximately 14 to 24 days on a 4000mAh 7.4V LiPo battery, making it feasible for deployment in remote agricultural fields. Future enhancements, such as the integration of solar panels or other energy harvesting technologies, could further extend the system's operational lifespan, potentially making the sensor nodes completely self-sufficient and maintenance free.

The project's results demonstrate minimal discrepancies between the sensor readings and actual environmental values, proving the system's accuracy and reliability. These findings validate the effectiveness of the chosen sensors, DHT22, HW-390, and BH1750 in providing precise data essential for precision agriculture.

In conclusion, this project offers a scalable, cost effective solution to the challenges of modern precision agriculture. The robust communication capabilities, low power consumption, and accurate data collection make this system a valuable tool for farmers looking to increase yields, conserve resources, and implement sustainable farming practices. With future advancements in energy harvesting, the system has the potential to become even more efficient, paving the way for a new era of smart agriculture.

References:

https://github.com/lamik/Light_Sensors_STM32/tree/master

<https://github.com/cloidnerux/STM32-DHT22/tree/master/mbed>

<https://os.mbed.com/platforms/ST-Nucleo-WL55JC/>

https://github.com/chmustafaj/Container_Analysis/tree/main

Appendices:

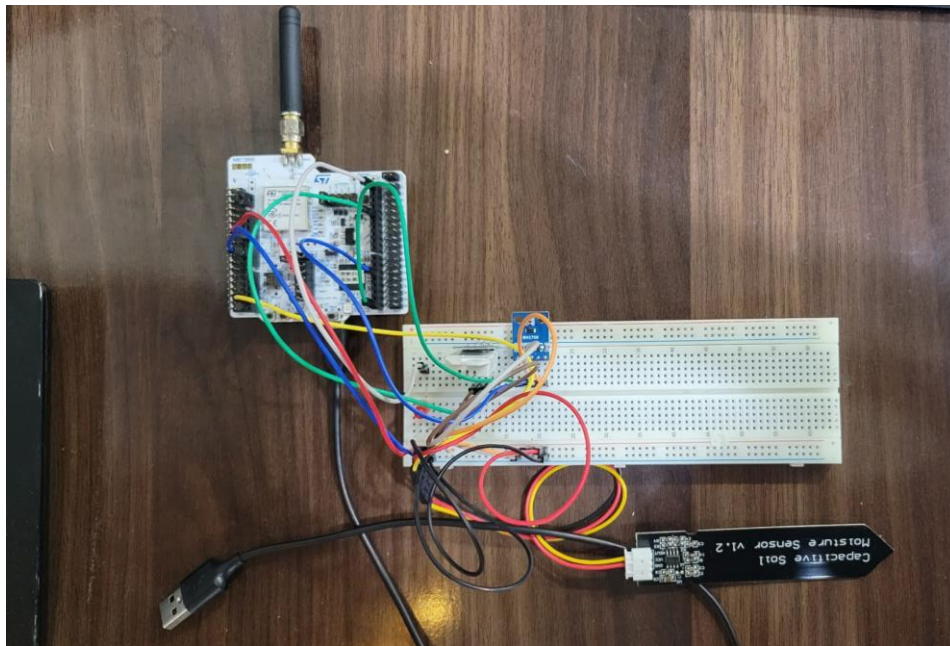


Figure 4: Complete Circuit on Breadboard

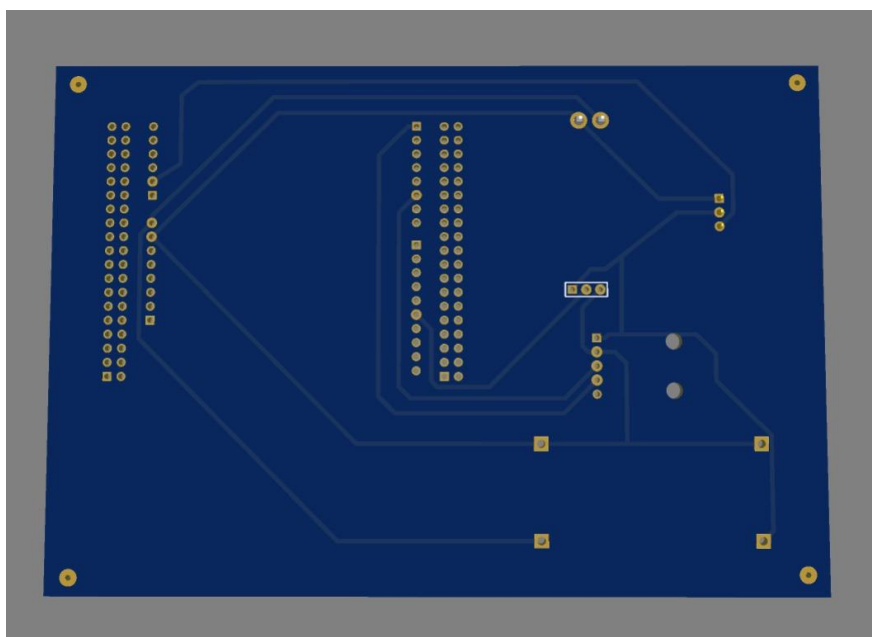


Figure 5: PCB Layout

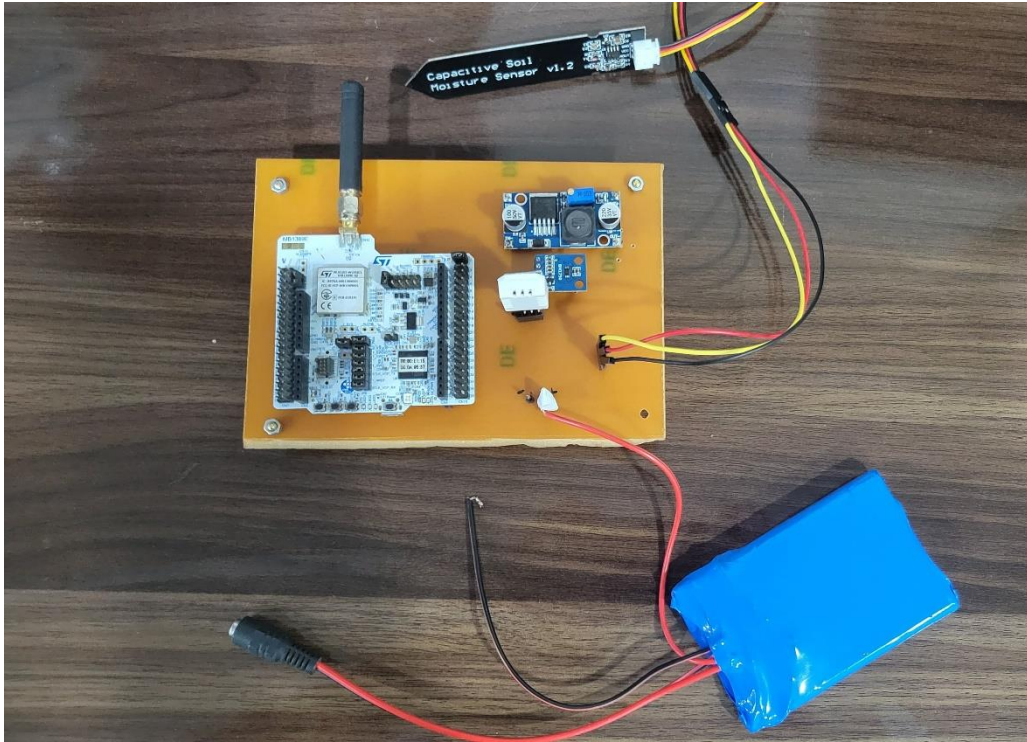


Figure 6: Actual PCB